Radicalization Kinetics under Algorithmic Exposure in a Stochastic Multiplex Model of Opinion Dynamics

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Abstract

We study how physical mobility, algorithmic exposure, and repulsive social influence interact in a stochastic multiplex model of opinion dynamics. Agents diffuse in physical space while a directed digital network rewires under a conserved attention budget, so digital exposure displaces rather than supplements local interaction. With purely assimilative bounded-confidence influence, opinion-blind long-range exposure reduces locality-induced fragmentation whereas homophilic recommendation preserves echo chambers. When a contested repulsive response to sufficiently distant opinions is activated, this ordering reverses at the reference parameters: a neutral platform reaches the maximal polarization permitted by the bounded opinion space, controversy-seeking curation drives faster initial separation but slows sharply near the boundary, and homophilic curation delays radicalization by suppressing cross-bloc exposure. In a late-stage symmetric two-bloc reduction, any curation kernel maps to a state-dependent cross-bloc exposure profile $p(y)$ and an exact quadrature for the radicalization time. Pointwise-ordered profiles inherit a global kinetic ordering; crossing profiles yield target- and horizon-dependent rankings. For similarity-driven curation the quadrature has a closed form involving the exponential integral. Simulations, finite-size scans to $N = 1600$, structural controls, and a well-mixed particle comparison support the mechanism. Heavy-tailed influence strengths are not required for the inversion; in the well-mixed heavy-tail regime they additionally produce a non-self-averaging stable-weighted asymptotic description. Finally, opinion-independent Brownian mobility produces no detectable geographic opinion structure in the explored regime, whereas opinion-dependent drift produces spatial domains through a Péclet-controlled crossover near $\chi\ell/D \sim 1$.

Keywords: opinion dynamics, sociophysics, complex networks, bounded confidence, algorithmic curation, stochastic processes, polarization

1. Introduction

The most common mechanistic story about platforms and polarization runs through similarity: recommendation algorithms feed users content they already agree with, creating echo chambers whose members drift apart from the rest of society [1–3]. The story suggests an obvious remedy—expose people to the other side. Yet the best-known field experiment on this intervention found the opposite of the intended effect: Republicans exposed for a month to a stream of liberal content became substantially more conservative [4] (the broader field record is mixed: other interventions have reduced polarization or left it unchanged [5, 6]). Social psychology offers a candidate mechanism for such backfire—contested, as we discuss below: influence is not always assimilative, and views far beyond an agent's latitude of acceptance can repel rather than attract [7–9]. That algorithmic mediation and cross-cutting exposure can either polarize or depolarize is by now well established [10]: attraction–repulsion models show that exposure can polarize intolerant populations [11, 12]; polarization can rise with increasing connectivity itself, without any suppression of interaction [13]; post-distribution and recommendation algorithms can promote or mitigate polarization depending on their strategy [14–16]; mild random "nudges" toward out-group content depolarize an echo-chambered population while excessive nudging radicalizes it [17]; and collaborative-filtering recommendations produce analytically tractable polarization phases [18]. What these works establish qualitatively—that the sign of an exposure intervention is not fixed—they do not provide a kinetic reduction that compares curation strategies on a common quantitative basis. That is the gap addressed here. Under a conserved attention budget, the late-stage two-bloc dynamics maps a platform's curation kernel to a state-dependent fraction of cross-bloc exposure in the repulsive zone. This exposure profile controls the outward drift and, through an exact quadrature, sets the finite-time radicalization scale—by which we mean, throughout, growth of the mean absolute opinion $\langle|x|\rangle$ toward the boundary of opinion space. Pointwise-ordered exposure profiles inherit a global kinetic ordering; when profiles cross, their comparison is instead target- and horizon-

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dependent and is determined by the full quadrature.

This paper builds a minimal model in which that question has a sharp answer. Agents carry continuous opinions and interact through two layers simultaneously: a physical layer, where encounters are structured by geography—agents diffuse through a two-dimensional world by Brownian motion and influence each other through a short-range bounded-confidence kernel [19–21]—and a digital layer, a directed attention graph whose links are continuously rewired by a platform according to an engagement kernel $E(\Delta)$: the probability that content at opinion distance Δ is shown. A finite attention budget couples the layers: the digital attention share λ displaces physical interaction rather than adding to it. On top of this multiplex substrate [22] we add the two psychologically motivated ingredients: an assimilation–indifference–repulsion influence function [7, 9, 23] and heavy-tailed influence strengths.

Our headline result is an inversion in the finite-time platform ranking. With purely assimilative influence, the model reproduces the standard picture—an opinion-blind platform acts like infinite-range mobility and heals fragmentation, while algorithmic homophily freezes fragmentation into spatially delocalized echo chambers. With the repulsive channel active at the reference attention share, all three designs develop outward radicalizing drift, but with sharply different finite-horizon outcomes: the neutral platform reaches the maximal polarization permitted by the bounded opinion space, controversy-seeking curation approaches a strong plateau, and strong algorithmic homophily produces the slowest separation because it suppresses cross-bloc encounters. The distinction is kinetic rather than a universal asymptotic ranking: a controversy kernel can generate the fastest initial separation while becoming slower than the other designs near the boundary as its exposure profile changes with opinion distance.

The inversion can be understood directly from the structure of the model, which we reduce analytically in three steps. In the late-stage symmetric two-bloc reduction, the stationary fraction of attention slots pointing across blocs, $p(y) = E(2y)/[E(0) + E(2y)]$ for blocs at $\pm y$, controls the radicalization rate $\dot{y} = 2\alpha_{\text{tot}}\lambda\eta\, p(y)\, y$. The resulting exposure profiles explain the distinct kinetics of the three kernels: the neutral profile is constant, similarity-driven exposure is increasingly suppressed as the blocs separate, and the controversy profile crosses the neutral baseline, producing very rapid early separation followed by strong finite-horizon slowing near the boundary. Within this deterministic reduction, radicalization has no threshold in the attention share—any $\lambda > 0$ with nonzero cross-bloc exposure produces outward drift—and the reduction yields, with no fitted parameters, an exact finite-horizon crossover by quadrature together with a compact local-rate estimate $\lambda_c^{(0)}(T)$ whose values span two orders of magnitude across designs, are consistent with the simulated onsets of the neutral and controversy-driven platforms, and predict that no boundary-reaching crossover exists in the physical range for the similarity-driven platform. And integrating the fast-rewiring, well-mixed counterpart of the model reproduces the neutral $\mathrm{Var}(x)$-versus-λ curve of the spatial agent-based model nearly quantitatively away from the steep onset region, and tracks the controversy curve from below with a gap that narrows as λ grows.

Two secondary results delimit the role of geography. First, in a phase diagram spanning four decades of mobility D and the full attention range λ, the radicalized region at high λ is independent of D: the dependence of the final state on mobility becomes weak once digital attention dominates the interaction budget. Second, under opinion-independent Brownian mobility we detect no geographic opinion structure at any mobility—a null result whose first-order part follows from translation invariance and that qualifies a common intuition about local echo chambers. Within this model, geographic opinion structure requires opinion–position coupling: adding a Schelling-type homophilic drift of strength χ [24, 25], we find that spatial opinion domains emerge through a Péclet-controlled crossover centered near $\chi\ell/D \sim 1$, where homophilic drift and diffusion become comparable over the interaction range ℓ.

The ingredients of the model each have a literature, recently systematised by Starnini et al. [10]: bounded confidence [19, 20, 26–28], mobile agents [21], coevolving networks [29], multiplex opinion formation [22], algorithmic bias and link recommendation [1, 2, 15], activity-driven echo chambers [3], stubborn agents [30], adaptive confidence [31], and coupled opinion–position dynamics [24, 25]. Closest in spirit is the post-distribution model of de Arruda et al. [14], in which an algorithm's distribution strategy determines whether polarization is promoted or mitigated under attractive and repulsive interactions; Bellina et al. [18] derive stationary polarization phases for collaborative-filtering recommendations, whereas our object is the kinetics of attraction–repulsion dynamics under conserved attention. The contribution here relative to that line is threefold: (i) a conserved-attention multiplex construction in which digital exposure displaces physical interaction rather than simply adding another channel; (ii) within the late-stage two-bloc reduction, a mapping from an arbitrary curation kernel to a state-dependent repulsive-exposure profile whose quadrature yields explicit finite-time radicalization scales; and (iii) a quantitative comparison principle for platform designs—direct when their exposure profiles are pointwise ordered and target- and horizon-dependent when the profiles cross—tested against the full spatial model and a well-mixed particle counterpart without fitting the onset data.

2. Model

2.1. State variables

Each of N agents carries a position $\mathbf{r}_i(t) \in [0, L)^2$ with periodic boundary conditions and an opinion $x_i(t) \in [-1, 1]$. Agent i may also carry an influence strength s_i and a stubbornness flag. The digital layer is a directed graph $A_{ij}(t) \in \{0, 1\}$, where $A_{ij} = 1$ means that the platform currently shows content by agent j to agent i. Each agent has a fixed number k of attention slots, $\sum_j A_{ij} = k$ for all i and t: attention is conserved, and new digital connections can only be acquired by dropping old ones.

2.2. Dynamics

Positions follow

$$\mathrm{d}\mathbf{r}_i = \chi\,\mathbf{f}_i\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}_i(t), \tag{1}$$

where D is the diffusion coefficient and $\mathbf{f}_i$ is an optional homophilic drift (Level 5 below; $\chi = 0$ elsewhere): the kernel-weighted mean of minimum-image unit vectors pointing toward compatible neighbours ($|x_j - x_i| < \epsilon$) and away from incompatible ones within the same spatial cutoff used below. The signed directional contributions are normalized by their total kernel weight, and the drift is set to zero when no contributing neighbour is present. Opinions obey the Itô equation

$$\begin{aligned}\mathrm{d}x_i = \alpha_i \frac{\sum_j K(r_{ij})\,B_\epsilon(x_j - x_i)\,(x_j - x_i)}{\sum_j K(r_{ij})\,B_\epsilon(x_j - x_i)}\,\mathrm{d}t \\ + \beta_i \frac{\sum_j A_{ij}\,s_j\,F(x_i, x_j)}{\sum_j A_{ij}\,s_j}\,\mathrm{d}t \ + \ \sigma_x\,\mathrm{d}B_i(t),\end{aligned} \tag{2}$$

where Euler–Maruyama updates are projected onto $[-1, 1]$ after each step (a reflecting-boundary control appears in Sec. 4.4), r_{ij} is the minimum-image distance between agents, sums run over $j \neq i$, and the physical drift is set to zero for agents with no compatible neighbour inside the kernel cutoff—a common event in the sub-percolation regime studied below. The physical kernel is a truncated Gaussian,

$$K(r) = e^{-r^2/2\ell^2}\,\mathbf{1}(r < 3\ell), \tag{3}$$

and $B_\epsilon(\Delta) = \mathbf{1}(|\Delta| < \epsilon)$ implements bounded confidence. The truncation is not cosmetic: because the drift is normalised, an untruncated Gaussian would let an agent with no nearby compatible peers average with the entire population at $O(1)$ rate through the exponential tail, silently removing the mobility scale from the problem.

The influence function has the assimilation–indifference–repulsion form

$$F(x_i, x_j) = \begin{cases} x_j - x_i, & |x_j - x_i| < \epsilon_1, \\ 0, & \epsilon_1 \leq |x_j - x_i| < \epsilon_2, \\ -\eta\,(x_j - x_i), & |x_j - x_i| \geq \epsilon_2, \end{cases} \tag{4}$$

with the repulsive branch ($\eta > 0$) active at Level 4; at Levels 2–3, F reduces to bounded confidence with threshold $\epsilon_1 = \epsilon$. This three-zone form descends from social-judgment-theory models [7, 23]. The empirical evidence for the repulsive branch is mixed: stochastic actor-oriented analyses of longitudinal network data support it [9], while controlled laboratory experiments find little evidence of negative shifts [32]; Level 4 should be read as exploring the consequences of this contested mechanism, not as assuming it settled.

2.3. Attention budget

The rates in Eq. (2) satisfy

$$\alpha_i = \alpha_{\mathrm{tot}}\,(1 - \lambda), \quad \beta_i = \alpha_{\mathrm{tot}}\,\lambda, \quad \lambda \in [0, 1], \tag{5}$$

so the digital attention share λ measures how much of a finite information-processing capacity is devoted to the platform. Increasing digital exposure displaces physical influence rather than adding to it, alongside the hard slot constraint $\sum_j A_{ij} = k$.

2.4. The platform: engagement-driven rewiring

In each interval $\mathrm{d}t$, agent i replaces one uniformly chosen attention slot with probability $\rho\,\mathrm{d}t$; the platform selects the replacement source j with probability

$$P(i \to j) \ \propto \ E\big(|x_i - x_j|\big)\ s_j, \tag{6}$$

where E is the platform's engagement kernel; the replacement is sampled among non-current sources, a finite-k correction of order k/N to the stationary exposure distribution used in Sec. 3, vanishing as $N \to \infty$. We compare three designs:

$$\begin{aligned} E_{\mathrm{sim}}(\Delta) = e^{-\gamma\Delta}, \qquad E_{\mathrm{neu}}(\Delta) = 1, \\ E_{\mathrm{con}}(\Delta) = e^{-(\Delta-\delta)^2/2w^2}. \end{aligned} \tag{7}$$

The similarity kernel models algorithmic homophily with strength γ [1, 2]; the neutral kernel is an opinion-blind baseline; the controversy kernel encodes a platform that maximises engagement by promoting content at a preferred ideological distance δ, close in spirit to the post-distribution algorithms of de Arruda et al. [14].

2.5. Model levels, heterogeneity, and observables

The model is built in five nested levels: (1) Brownian motion + bounded confidence; (2) static opinion-neutral digital links; (3) adaptive similarity-driven rewiring; (4) general engagement kernels + repulsion + heavy-tailed influence strengths (optional stubborn agents [30]); (5) homophilic mobility $\chi > 0$ added to the purely physical model of Level 1. Influence strengths are drawn as $s_i = 1 + z_i$ with z_i Lomax-distributed—density tail $p(s) \sim s^{-\kappa}$, i.e. survival exponent $\kappa - 1$—and normalised to unit mean per realisation. At the reference $\kappa = 2.5$ the variance of s is infinite: a deliberate "influencer" feature whose finite-N fluctuations play a visible role in Sec. 4.5.

We measure: polarization $P = \mathrm{Var}(x)$; extremism $\langle |x| \rangle$; the number of opinion clusters n_c (gaps larger than 0.05 in sorted opinion space, groups of at least two agents); a categorical state (consensus / polarization / fragmentation for $n_c \leq 1$ / $= 2$ / ≥ 3; states with $\mathrm{Var}(x) < 10^{-2}$ are additionally classed as consensus, an override that never fires in the reported data; note the scalar P and the two-cluster categorical state share the word "polarization"); spatial autocorrelation of opinions via Moran's I (with Gaussian spatial weights of scale ℓ) and via the local agreement gap

$\Pr(|x_i - x_j| < \epsilon \mid r_{ij} < 3\ell) - \Pr(|x_i - x_j| < \epsilon)$; opinion assortativity ρ of the digital graph; modularity Q of the digital graph with respect to opinion clusters; and mean local disagreement $\langle |x_i - \bar{x}_{\partial i}| \rangle$ between an agent and the average opinion it is shown.

2.6. Numerical integration

Equations (1)–(2) are integrated by Euler–Maruyama. Unless stated otherwise, $N = 200$, $L = 1$, $\mathrm{d}t = 0.02$, $\alpha_{\rm tot} = 1$, $\epsilon = \epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$, $\ell = 0.02$, $k = 10$, $\rho = 5$, $\gamma = 4$, $\delta = 0.8$, $w = 0.2$, and $\kappa = 2.5$. Opinion noise is $\sigma_x = 0.02$ whenever the digital layer is adaptive (Levels 3–4, including the phase diagram) and $\sigma_x = 0$ in the purely physical and static-layer experiments (Levels 1, 2, and 5). Horizons are $T = 60$ (Levels 1–3 and 5), $T = 80$ (Level 4, the robustness scans, and the attention sweeps), and $T = 50$ (the phase diagram). In system-size scans the box is rescaled as $L = \sqrt{N/200}$ so that density, kernel range and digital degree stay constant. With $\ell = 0.02$ the mean physical contact degree is $N\pi(3\ell)^2/L^2 \approx 2.3$, below the percolation threshold of the random geometric graph, so the instantaneous contact network is spatially fragmented and locality is meaningful. Statistics use 6–12 realisations per point (3 for the phase diagram); error bars denote the standard deviation across realisations. Halving $\mathrm{d}t$ twice shifts the Level-4 polarization values by amounts comparable to that spread ($\lesssim 0.05$, concentrated in the similarity platform and directed upward—i.e. against the inversion, which is therefore conservative) without affecting the platform ordering.

3. Theory: two-bloc reduction and well-mixed counterpart

3.1. Late stage: cross-bloc attention controls radicalization

Consider the late-stage configuration observed at Level 4: two equal blocs at opinions $\pm y$. When attention slots renew quickly compared with the bloc drift (slot renewal rate ρ/k against the rate $2\alpha_{\rm tot}\lambda\eta p$ of Eq. (9) below—a condition that is only marginally met for the neutral kernel at our reference parameters, so the agreement found below also attests to the reduction's robustness against partial slot memory), the stationary probability that an attention slot of a +-bloc agent points into the − bloc follows from Eq. (6):

$$p(y) = \frac{E(2y)}{E(0) + E(2y)}. \tag{8}$$

(Equation (8) is exact in the mean for homogeneous strengths $s_i = 1$, which is therefore the natural theoretical baseline; a homogeneous-strength control reproduces the inversion unchanged (Sec. 4.4). With heavy-tailed strengths, slots are both sampled and weighted proportionally to s, and the realised cross-bloc exposure fluctuates around the homogeneous-strength prediction: for a realised population the slot-sampling probability is $E(2y)S_\mp/[E(0)S_\pm + E(2y)S_\mp]$, with $S_\pm$ the two blocs' realised strength totals, which reduces to Eq. (8) as their ratio approaches unity—guaranteed by the finite mean of the strength law, though convergence with N is slow for our infinite-variance reference distribution. Given the slot types, the expected s-weighted drift then equals the unweighted one by slot exchangeability, since strengths are independent of bloc membership; see Sec. 3.3.) Within-bloc content exerts no drift ($F = 0$ at $\Delta = 0$), and cross-bloc content is repulsive once $2y \geq \epsilon_2$, so the bloc positions obey

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2\,\alpha_{\rm tot}\,\lambda\,\eta\; p(y)\; y, \qquad 2y \geq \epsilon_2, \tag{9}$$

with no physical restoring force: once the blocs are separated beyond the confidence bound, the physical layer only pulls agents toward their own bloc's mean. Equation (9) is the central statement of this paper within the late-stage symmetric two-bloc reduction: any curation kernel E is mapped to a state-dependent cross-bloc exposure profile $p(y)$, whose quadrature sets the radicalization time [Eq. (10) below]. If two profiles are pointwise ordered over the traversed opinion range, their radicalization times inherit the opposite ordering directly. If the profiles cross, no single global ranking exists independently of the target and observation horizon; the full quadrature is then required. The three kernels of Eq. (7) illustrate both cases: neutral and similarity are pointwise ordered, whereas the controversy profile crosses the neutral baseline as the blocs separate. Their distinct $p(y)$ profiles therefore produce the following kinetics:

- Neutral: $p \equiv 1/2$. The drift $\alpha_{\rm tot}\lambda\eta y$ never shuts off; opinions are driven to the boundary and pinned at ± 1.
- Controversy-driven: $p(y) \approx 1$ for $2y \gtrsim \epsilon_2$ (cross-bloc content is exactly what engages), so separation is fast. For the Gaussian kernel of Eq. (7), $p_{\rm con}(y) = 1/2$ at $y = \delta$ (besides the trivial $y = 0$ solution), so the controversy profile crosses the neutral baseline there; for $y > \delta$ its cross-bloc exposure is smaller than neutral. As $2y \to 2$, both $E(2y)$ and $E(0)$ are far from the engagement peak and $p \to E(2)/[E(0) + E(2)] \approx e^{2(\delta-1)/w^2}$, which for $\delta < 1$ is exponentially small. The dynamics therefore becomes exponentially slow near the boundary, producing an apparent finite-horizon plateau: the platform continues to exert a nonzero outward drift, but its cross-bloc exposure becomes so small that further motion is negligible on the simulated horizons.
- Similarity-driven: $p(y) = e^{-2\gamma y}/(1 + e^{-2\gamma y})$ is small for $\gamma y \gtrsim 1$. Radicalization proceeds, but at a rate suppressed by $e^{-2\gamma y}$: a slow outward creep fed by the few cross-cutting links that stochastic rewiring keeps producing.

These three behaviours are exactly what the simulations show (Sec. 4.3): pinning at ± 1 for neutral ($\langle |x| \rangle \approx 1.00$), an apparent plateau short of the boundary for controversy (≈ 0.95), and slow partial radicalization for similarity (≈ 0.79 at $T = 80$). Note also that within the deterministic reduction Eq. (9) has no positive critical λ: any $\lambda > 0$ with nonzero cross-bloc exposure produces outward drift. What varies across designs—by orders of magnitude—is the rate, which we quantify next.

3.2. Onset: radicalization is rate-limited, not threshold-limited

One might expect a critical attention share below which the physical layer's assimilative pull protects the population. A deliberately naïve stability estimate around the initially uniform state gives such thresholds: for an edge agent, the bounded-confidence physical term has inward magnitude $(1-\lambda)\epsilon/2$, while the digital term is the engagement-weighted repulsive drift generated by opinions beyond ϵ_2; equating these two contributions at the reference parameters gives $\lambda^* \approx 0.34$ for the neutral kernel and 0.55 for the controversy kernel. These values are used only as a foil—the simulations show radicalizing drift already at $\lambda \approx 0.05$ (Sec. 4.5). The failure is mechanistic rather than numerical: bounded-confidence fragmentation moves the system away from the uniform state before that putative instability can control the late-time dynamics. For attention shares bounded away from $\lambda = 1$, bounded-confidence fragmentation forms first on the timescale $\sim (1-\lambda)^{-1}$, producing blocs near $\pm y_0$ with $y_0 \approx 0.6$ (the bounded-confidence cluster positions at $\epsilon = 0.3$, read off from Figs. 1 and 3, not fitted to the threshold data). Since $2y_0 > \epsilon_2$, this fragmented state is unstable to repulsion for any $\lambda > 0$: once separated beyond the confidence bound, the physical layer exerts no restoring force between blocs (Sec. 3.1). The timescale separation necessarily weakens as $\lambda \to 1$, where the physical layer vanishes; in that limit y_0 should be interpreted as a late-stage initialization of the two-bloc reduction rather than as the outcome of a parametrically faster physical transient.

What remains is a rate. Equation (9) is separable, so the time to traverse from y_0 to y_f follows exactly by quadrature,

$$t(y_0 \to y_f) = \frac{1}{2\,\alpha_{\text{tot}}\,\lambda\,\eta} \int_{y_0}^{y_f} \frac{\mathrm{d}y}{y\,p(y)}, \tag{10}$$

which maps any curation kernel to a radicalization timescale through its exposure profile alone. Because $t \propto 1/\lambda$, the exact finite-horizon crossover for reaching a target y_f within an observation window T follows immediately:

$$\lambda_c(T; y_f) = \frac{1}{2\,\alpha_{\text{tot}}\,\eta\,T} \int_{y_0}^{y_f} \frac{\mathrm{d}y}{y\,p(y)}, \tag{11}$$

whenever the right-hand side does not exceed one; otherwise the target is unreachable within the window at any attention share. For the similarity kernel, the quadrature is available in closed form because $p_{\text{sim}}(y) = 1/[1+\exp(2\gamma y)]$:

$$t_{\text{sim}}(y_0 \to y_f) = \frac{\ln(y_f/y_0) + \mathrm{Ei}(2\gamma y_f) - \mathrm{Ei}(2\gamma y_0)}{2\,\alpha_{\text{tot}}\,\lambda\,\eta}, \tag{12}$$

where Ei is the exponential integral. At the reference parameters with $y_0 = 0.6$ and $y_f = 1$, Eq. (12) gives $t_{\text{sim}} \simeq 507.7/\lambda$, so Eq. (11) gives $\lambda_c = 6.35$: the boundary cannot be reached within $T = 80$ even at $\lambda = 1$, and no boundary-reaching crossover exists in the physical interval $[0,1]$. For the controversy kernel the quadrature is evaluated numerically: from $y_0 = 0.6$, $t(0.6 \to 0.90) \simeq 3.2/\lambda$, $t(0.6 \to 0.95) \simeq 32.1/\lambda$, and $t(0.6 \to 1) \simeq 482/\lambda$. Fast initial radicalization is followed by a dramatic slowing: the plateau level $\langle |x| \rangle \approx 0.95$ observed in the simulations is reached within $T = 80$ only for $\lambda \gtrsim \lambda_c(80; 0.95) \approx 0.40$, while the boundary itself lies far beyond the principal $T = 80$ observation window and is not reached on the simulated horizons.

For a compact local-rate estimate applicable to all three designs, we also use the constant-exposure approximation $p(y) \approx p_0 = p(y_0)$, which gives the local timescale

$$t_{\text{loc}}(\lambda) = \frac{\ln(y_f/y_0)}{2\,\alpha_{\text{tot}}\,\lambda\,\eta\,p_0}, \tag{13}$$

and the corresponding local-rate crossover estimate $t_{\text{loc}} = T$:

$$\lambda_c^{(0)}(T) = \frac{\ln(y_f/y_0)}{2\,\alpha_{\text{tot}}\,\eta\,p_0\,T}. \tag{14}$$

The frozen-p_0 approximation is exact for the neutral kernel, whose $p \equiv 1/2$ is state-independent, so there $\lambda_c^{(0)}$ coincides with Eq. (11); for the state-dependent kernels it characterizes the initial rate only. Table 1 evaluates Eqs. (13)–(14) at the reference parameters with no fitting. The cross-bloc slot fraction p_0 spans two orders of magnitude across the designs, and with it the local rate: for the neutral platform $\lambda_c^{(0)} = \lambda_c \approx 0.016$ and for the controversy platform $\lambda_c^{(0)} \approx 0.008$—any noticeable digital attention starts radicalization within the horizon, though for controversy the exact plateau-reaching crossover is the larger $\lambda_c(80; 0.95) \approx 0.40$ above. For the similarity platform $\lambda_c^{(0)} \approx 0.98$ is a local-rate figure only: the exact quadrature shows that no boundary-reaching crossover exists in the physical range, and even full digital attention only partially radicalizes the population in the same window—exactly the smooth partial growth observed in the simulations. "Protection" by homophilic curation is therefore kinetic, not asymptotic. At the initial two-bloc state its local rate is suppressed by the factor p_0 relative to maximal cross-bloc exposure; relative to the neutral baseline $p = 1/2$, the local timescale is enhanced by $(2p_0)^{-1}$. Because $p_{\text{sim}}(y)$ decreases further as the blocs separate, the exact quadrature gives an even larger finite-time delay without changing the destination within the deterministic reduction.

Two caveats delimit Table 1. The constant-p_0 approximation is exact for the neutral kernel but conservative for the similarity kernel, whose $p(y)$ decays further as the blocs separate: the exact time $507.7/\lambda$ is more than sixfold longer than $78.2/\lambda$, so the true kinetic protection is even stronger than the tabulated estimate suggests. For

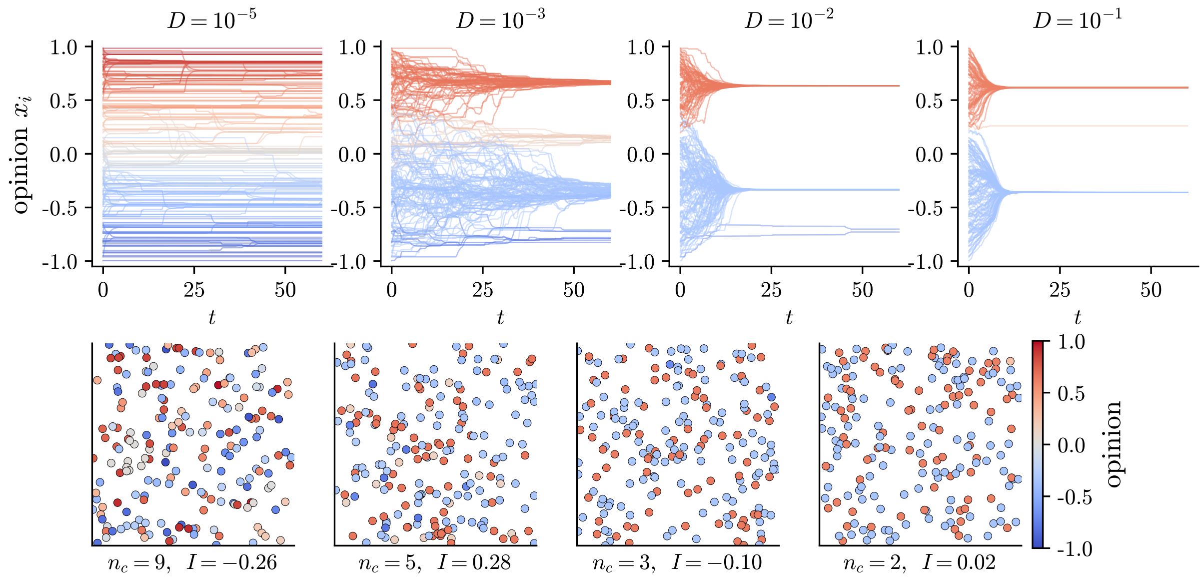

Figure 1: Purely physical dynamics (Level 1). Top: opinion trajectories $x_i(t)$ for increasing diffusion coefficient D; colours encode final opinions. Bottom: final spatial configurations. Panel captions report the cluster count n_c and Moran's I of the single realisation shown; individual-run values of I fluctuate around a zero ensemble mean (s.d. ≈ 0.13 across seeds, single-run extremes up to $|I| \approx 0.35$; Fig. 2). Low mobility freezes many coexisting clusters; high mobility recovers the mean-field bounded-confidence outcome. Note the absence of spatial colour domains at any D.

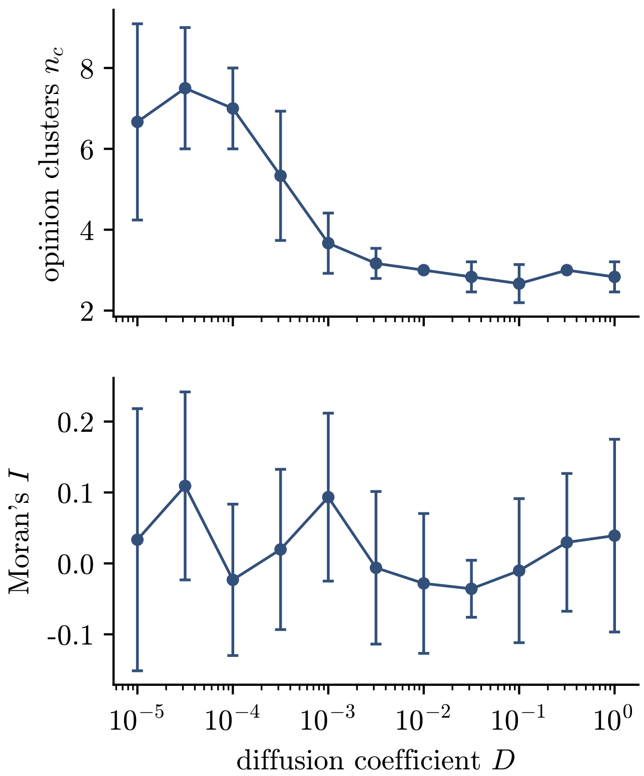

Figure 2: Level-1 sweep over the diffusion coefficient. Top: mean number of surviving opinion clusters. Bottom: spatial autocorrelation of opinions measured by Moran's I (mean ± s.d. over 6 realisations). Mobility interpolates between local fragmentation and mean-field behaviour while spatial opinion autocorrelation stays at noise level throughout.

Table 1: Two-bloc local-rate analysis, Eqs. (8) and (13)–(14), at the reference parameters ($y_0 = 0.6$, $y_f = 1$, $\eta = 0.4$, $\gamma = 4$, $\delta = 0.8$, $w = 0.2$, $T = 80$, $\alpha_{\text{tot}} = 1$). No parameters are fitted to these data; y_0 is the observed bounded-confidence cluster position. The tabulated $\lambda_c^{(0)}$ are frozen-p_0 estimates, exact only for the neutral kernel; exact target-crossing crossovers follow from Eq. (11) and are quoted in Sec. 3.2.

Platform	p_0	$t_{\text{loc}}(\lambda)$	$\lambda_c^{(0)}(T{=}80)$
Similarity-driven	0.0082	$78.2/\lambda$	0.98
Neutral	0.500	$1.28/\lambda$	0.016
Controversy-driven	0.998	$0.64/\lambda$	0.008

the controversy kernel, $p(y)$ collapses as $2y \to 2$ (Sec. 3.1), so t_{loc} measures the fast initial separation rate rather than the time to any fixed target; target-crossing times follow from Eq. (10).

3.3. Well-mixed limit

In the fast-rewiring, well-mixed counterpart of the model, each agent feels the global bounded-confidence mean with weight $1-\lambda$ and a population-wide engagement-weighted influence average with weight λ,

$$\mathrm{d}x = (1-\lambda)\,\frac{\mathbb{E}[(x'-x)\,\mathbf{1}(|x'-x|<\epsilon)]}{\mathbb{P}(|x'-x|<\epsilon)}\,\mathrm{d}t + \lambda\,\frac{\sum_j s_j^2\,E(|x_j-x|)\,F(x,x_j)}{\sum_j s_j^2\,E(|x_j-x|)}\,\mathrm{d}t + \sigma_x\,\mathrm{d}B, \tag{15}$$

where x' is an independent draw from the population law and the digital sum runs over the whole population. The squared strengths arise because slots are both sampled

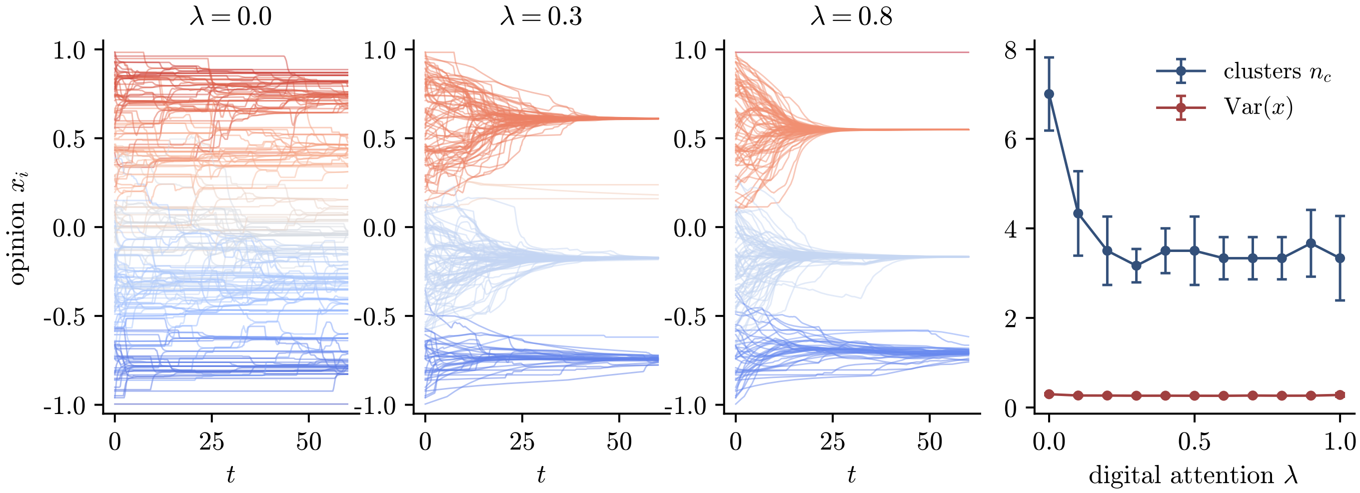

Figure 3: Level 2: static random digital layer at low mobility ($D = 10^{-4}$). Left three panels: opinion trajectories for increasing digital attention share λ. Right: cluster count and polarization versus λ (mean ± s.d.).

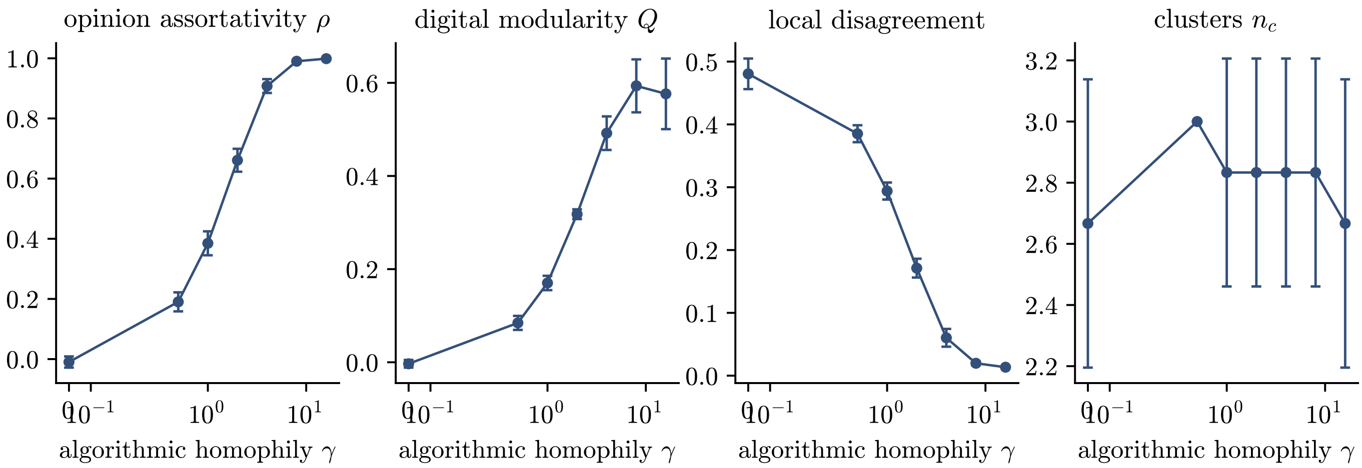

Figure 4: Level 3: adaptive digital layer with similarity-driven recommendation, $\lambda = 0.5$, $D = 10^{-3}$. From left to right: opinion assortativity of the digital graph, modularity with respect to opinion clusters, mean local disagreement, and cluster count as functions of the algorithmic homophily γ (mean ± s.d. over 6 runs).

and weighted proportionally to s. The status of Eq. (15) as $N \to \infty$ depends on the tail of the strength law. For $\kappa > 3$, $\mathbb{E}[s^2]$ is finite, the self-normalised digital drift self-averages, strengths drop out entirely (they are independent of opinions), and Eq. (15) converges to a deterministic McKean–Vlasov dynamics. At our reference $\kappa = 2.5$, however, $\mathbb{E}[s^2] = \infty$ and the size-biased slot-strength law has infinite mean, so the digital drift does not self-average: it remains a random variable dominated by the few largest strengths as population size grows—influencer fluctuations therefore do not vanish through conventional self-averaging in this reference heavy-tail regime. For the finite-mean regime $\kappa > 2$, the slot-level two-bloc expectation in Sec. 3.1 remains strength-free because slot types are independent of the selected strengths. The population-wide limit in Eq. (15) is different when $2 < \kappa < 3$. Writing $Z = s^2$, its survival tail is regularly varying with index $\alpha = (\kappa-1)/2 \in (1/2, 1)$. Under the standard marked regular-variation approximation, treating the strength marks as asymptotically independent of bloc membership, the normalized sums in Eq. (15) converge in distribution to positive α-stable random measures rather than to their expectations [33]. Multiplying the regularly varying weight Z by the bounded positive engagement mark E rescales its tail intensity by E^α, so the corresponding stable control measure is proportional to E^α. In this asymptotic description the well-mixed digital drift remains random, while its annealed mark weighting is proportional to E^α rather than E. In the two-bloc notation, the corresponding annealed cross-bloc fraction is

$$p_\alpha(y) = \frac{E(2y)^\alpha}{E(0)^\alpha + E(2y)^\alpha}, \qquad \alpha = \frac{\kappa - 1}{2}, \tag{16}$$

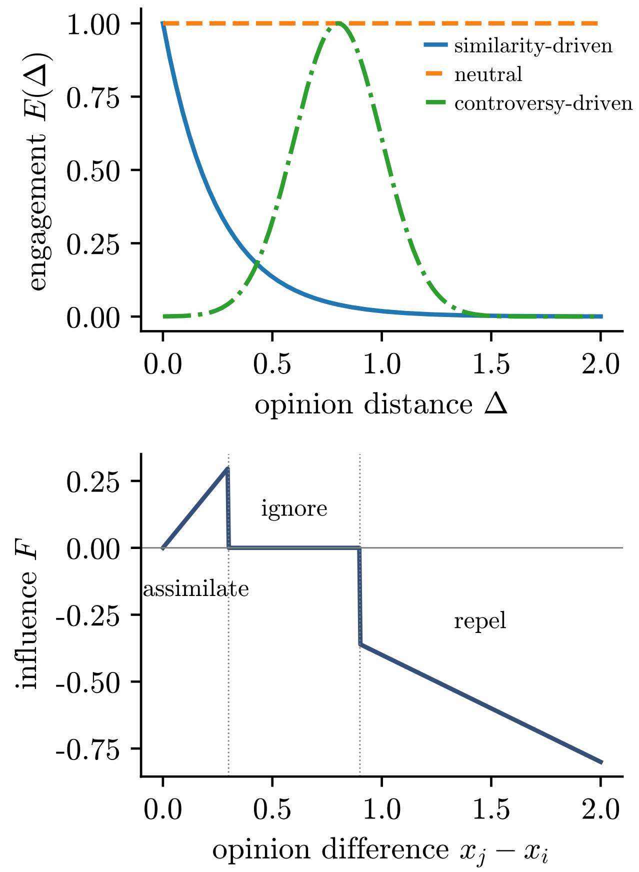

Figure 5: Level-4 ingredients. Top: the three engagement kernels of Eq. (7) ($\gamma = 4$, $\delta = 0.8$, $w = 0.2$). Bottom: the assimilation–indifference–repulsion influence function of Eq. (4) ($\epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$).

which reduces to the ordinary deterministic weighting only when the second moment becomes finite. At the reference $\kappa = 2.5$, $\alpha = 3/4$: heavy-tailed influencer fluctuations therefore flatten algorithmic selectivity in the population-wide annealed limit while remaining non-self-averaging realization by realization. This distinction does not underwrite the inversion itself, which persists for $s_i = 1$ and $\kappa = 4$ (Sec. 4.4); it clarifies instead why the heavy-tailed well-mixed comparison is a stochastic limit rather than a deterministic McKean–Vlasov closure.

We therefore compare the spatial agent-based model against a direct integration of Eq. (15) as a spaceless well-mixed particle model ($M = 600$, 12 realisations, same strength law), i.e. the model stripped of space and of the finite attention window, in Sec. 4.5: with no fitted parameters it reproduces the neutral curve nearly quantitatively away from the steep onset region, tracks the controversy-driven curve from below with a gap that narrows as λ grows, and tracks the shape of the similarity-driven curve, which it systematically underestimates because the slot-level sampling noise it discards (each spatial agent draws only $k = 10$ sources) is what feeds the similarity platform's slow creep. The inversion is thus a property of the opinion dynamics and the engagement weighting, not an artefact of space or slow rewiring.

4. Results

4.1. Baselines: mobility and neutral connectivity (Levels 1–2)

Figures 1–3 establish the assimilative baselines. In the purely physical model, the competition between the local convergence time $\tau_{\rm op} \sim \alpha_{\rm tot}^{-1}$ and the neighbourhood-renewal time $\tau_\ell \sim \ell^2/D$ controls how many opinion clusters survive: $n_c \approx 7 \pm 2$ frozen local clusters for $D \lesssim 10^{-4}$, merging into the mean-field cluster set ($n_c \approx 2$–3 at $\epsilon = 0.3$ [19]) for $D \gtrsim 10^{-2}$, consistent with mobile-agent simulations [21]. Adding a static, opinion-blind digital layer at low mobility heals this excess fragmentation: a modest attention share ($\lambda \approx 0.2$–0.3) already merges the frozen clusters down to $n_c \approx 3.3$–3.5, close to the mean-field set, beyond which more digital attention changes nothing (Fig. 3). Opinion-blind long-range exposure acts, in effect, like infinite-range mobility. (These levels run at $\sigma_x = 0$; repeating the Level-1 sweep at the adaptive levels' noise $\sigma_x = 0.02$ reduces the frozen cluster count—noise merges micro-clusters, $n_c = 4.8 \pm 1.6$ at $D = 10^{-5}$—but leaves the fragmentation-to-mean-field crossover intact, so the level comparison is not a noise artefact.) In the assimilative world, connectivity is benign; everything that follows is about what curation and repulsion do to this baseline.

4.2. Algorithmic homophily builds delocalised echo chambers (Level 3)

With similarity-driven rewiring (still purely assimilative influence), the feedback loop opinion → topology → exposure → opinion closes as γ grows (Fig. 4): opinion assortativity climbs to one, digital modularity rises, and the disagreement visible in an agent's feed collapses while global fragmentation persists. This is an echo chamber in the precise sense of Baumann et al. [3]—high global variance, near-zero visible disagreement—via the mechanism of Sîrbu et al. [1] and Santos et al. [2] transplanted into a coevolving multiplex [29]. The communities are spatially delocalised: geography carries no trace of them. In the assimilative world this is the worst a platform can do in our model: freeze moderate fragmentation and hide it from its participants.

4.3. The inversion: uncurated exposure radicalizes (Level 4)

Activating the repulsive branch and heavy-tailed influence strengths (Fig. 5) produces the inversion (Fig. 6). All three designs end with two antagonistic blocs, but with sharply different degree and speed, in the order predicted by the two-bloc reduction of Sec. 3.1. The neutral platform is the most polarizing ($\mathrm{Var}(x) = 0.98 \pm 0.03$ over 24 realisations, opinions pinned at ± 1 within a few time units):

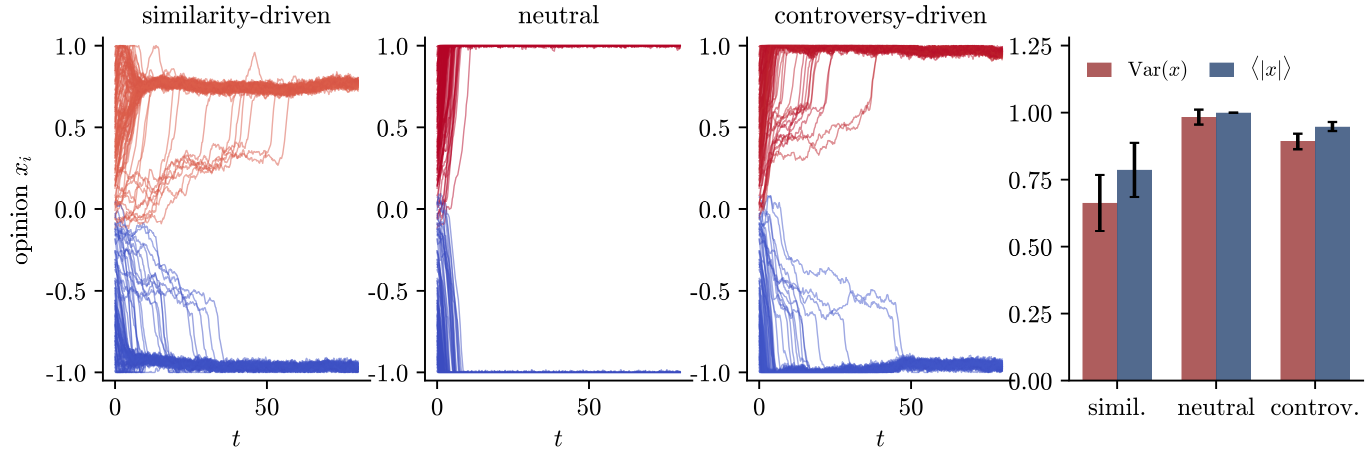

Figure 6: Level 4: platform designs compared under repulsive influence and heavy-tailed influence strengths ($\lambda = 0.6$). Left three panels: representative opinion trajectories. Right: polarization Var(x) and extremism $\langle|x|\rangle$ (mean $\pm$ s.d. over 24 runs).

with $p = 1/2$, half of every feed is repulsion-triggering at all times. The controversy platform is nearly as extreme (0.89 ± 0.03) but plateaus just short of the boundary on the simulated horizon ($\langle|x|\rangle \approx 0.95$), as predicted by the collapse of $p(y)$ when $2y \to 2$. The similarity platform is the least polarizing (0.66 ± 0.10): hiding opposed content starves the repulsive channel, leaving only the slow creep fed by residual cross-cutting links. Because the influence strengths have infinite variance, we also checked the distribution shape: medians (0.99/0.90/0.68) and interquartile ranges give the same ordering as the means, with no overlap between platforms. The network signatures decouple from the opinion signatures: the neutral platform reaches maximal polarization with zero opinion assortativity ($\rho \approx 0$)—polarization without echo chambers—whereas both curated platforms end almost perfectly assortative ($\rho \approx 0.9$–1.0).

4.4. Robustness: repulsive-zone exposure orders the designs

Figure 7 tests the ordering across the repulsion and curation parameters. Across the full (η, ϵ_2) grid the order neutral $\geq$ controversy $>$ similarity holds in the mean at every grid point, with separations much larger than the across-seed spread at the widest repulsive zone ($\epsilon_2 = 0.7$); the gaps narrow as ϵ_2 grows until the controversy–similarity gap falls within the seed spread at $\epsilon_2 = 1.3$; all platforms polarize less as ϵ_2 grows (repulsion becomes harder to trigger) and more as η grows. The two curated platforms interpolate toward the neutral level in exactly the way the theory predicts. The similarity platform approaches the neutral bound as γ decreases toward zero (curation too weak to withhold repulsive content)—already at $\gamma = 1$, the weakest curation simulated, final polarization lies close to the neutral reference—and its protective effect saturates for $\gamma \gtrsim 8$. The controversy platform sits below the neutral bound while its engagement peak $\delta < \epsilon_2$ (much of its exposure lands in the indifference band) and reaches the bound once $\delta \geq \epsilon_2$, when engagement maximisation and repulsion maximisation coincide—at which point it matches the neutral end state while initially radicalizing at twice the neutral rate (Table 1). The neutral platform thus realizes the maximal polarization permitted by the bounded opinion space, and the curated designs approach this level as their exposure profiles place increasing weight in the repulsive zone.

Figure 8 scans the system size at constant density up to $N = 1600$. The three platforms' polarization and extremism are essentially flat in N: the inversion persists across the tested system sizes and is not a small-N artefact.

Table 2 adds five further controls at the reference point ($\lambda = 0.6$, 12 realisations each). The inversion survives, with the same ordering: (i) homogeneous strengths $s_i = 1$—the regime in which the two-bloc reduction is exact in the mean—so neither the heavy tail nor influence heterogeneity is needed for the effect; (ii) a finite-variance strength law ($\kappa = 4$); (iii) reflecting instead of clipping boundaries, so the saturation ordering is not an artefact of the projection; (iv) halved and doubled attention windows ($k = 5, 20$), which shift the similarity platform's creep in the direction predicted by the slot-fluctuation mechanism (fewer slots, fewer cross-bloc bursts); and (v) slower and faster rewiring ($\rho = 1, 20$). The last is the most informative: fast rewiring strengthens homophilic protection (Var = 0.54 at $\rho = 20$) while slow rewiring weakens it (0.87 at $\rho = 1$, where the controversy–similarity gap narrows into the seed spread)—exactly the dependence expected from the quasi-stationarity assumption behind Eq. (8), since slowly renewed feeds let occasional cross-bloc links persist and act for longer.

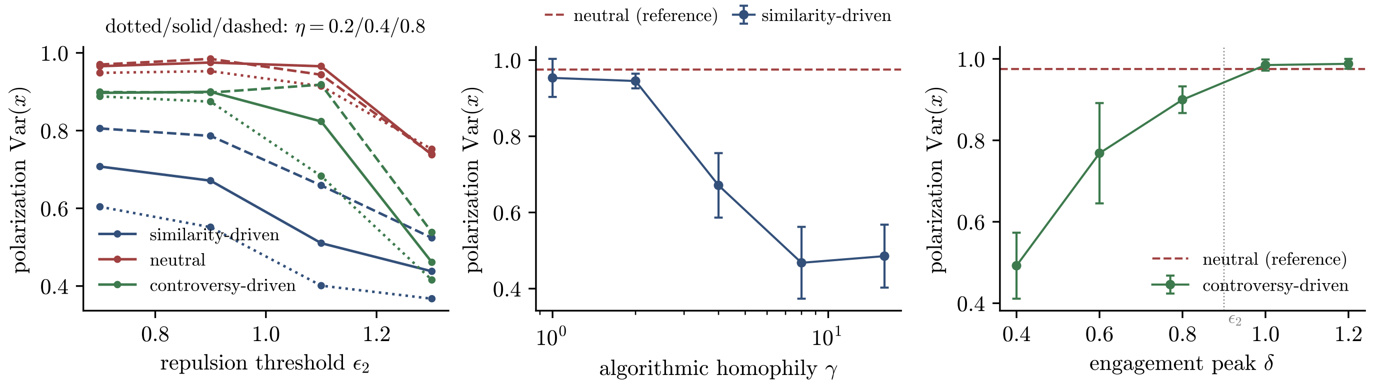

Figure 7: Robustness of the inversion. Left: final polarization versus the repulsion threshold ϵ_2 for $\eta = 0.2/0.4/0.8$ (dotted/solid/dashed); colours denote platforms (blue similarity, red neutral, green controversy). Centre: similarity platform versus algorithmic homophily γ, with the neutral platform as reference. Right: controversy platform versus engagement peak δ; the dotted line marks $\delta = \epsilon_2$. Mean ± s.d. over 6 runs.

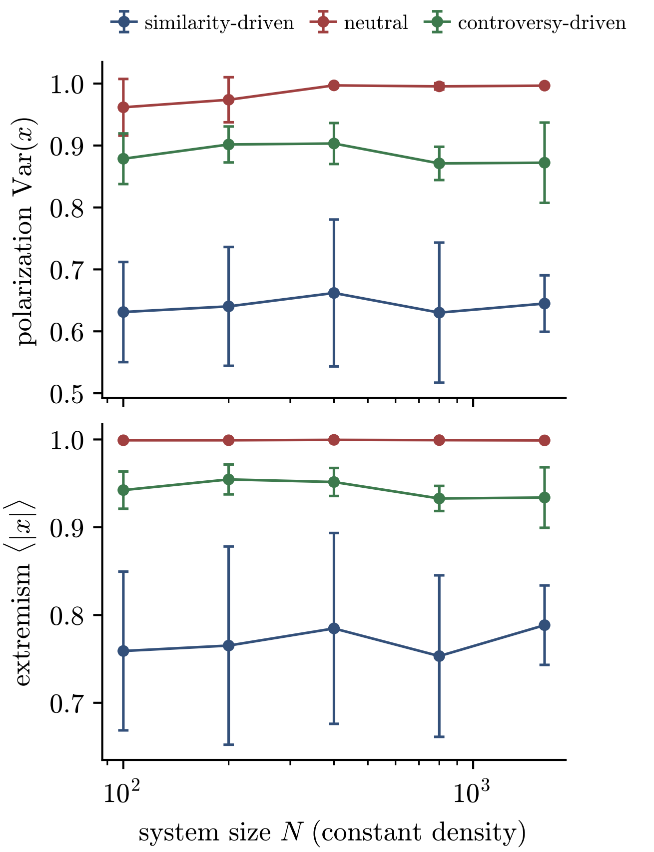

Figure 8: System-size dependence of the inversion at constant density ($L = \sqrt{N/200}$), $\lambda = 0.6$. Polarization (top) and extremism (bottom) for the three platforms, $N = 100$–1600.

Table 2: Robustness controls at the reference point ($\lambda = 0.6$, $T = 80$): final polarization Var(x), mean ± s.d. over 12 realisations. The reference row uses 24 realisations.

Variant	Similarity	Neutral	Controversy
Reference	0.66 ± 0.10	0.98 ± 0.03	0.89 ± 0.03
$s_i = 1$	0.68 ± 0.08	1.00 ± 0.01	0.91 ± 0.02
$\kappa = 4$	0.67 ± 0.09	0.99 ± 0.01	0.91 ± 0.02
Reflecting	0.62 ± 0.12	0.97 ± 0.04	0.89 ± 0.03
$k = 5$	0.54 ± 0.08	0.98 ± 0.02	0.90 ± 0.03
$k = 20$	0.76 ± 0.11	0.98 ± 0.03	0.92 ± 0.03
$\rho = 1$	0.87 ± 0.08	0.98 ± 0.02	0.91 ± 0.03
$\rho = 20$	0.54 ± 0.10	0.97 ± 0.04	0.88 ± 0.05

4.5. The radicalization crossover and the well-mixed model

Figure 9 compares the simulated Var(x)–λ curves with the theory of Sec. 3. The neutral platform jumps to its saturated value already at $\lambda = 0.05$–0.10, the finest nonzero attention shares sampled—consistent with its predicted crossover $\lambda_c \approx 0.016$, which lies below the grid resolution, and inconsistent with the naive uniform-state threshold (0.34). The controversy platform starts radicalizing equally early ($\lambda_c^{(0)} \approx 0.008$) but rises gradually, from Var(x) ≈ 0.72 at $\lambda = 0.05$ to its ≈ 0.90 plateau around $\lambda \approx 0.35$–0.40—in agreement with the exact quadrature, which puts the plateau-reaching crossover at $\lambda_c(80; 0.95) \approx 0.40$ (Sec. 3.2), and inconsistent with the naive uniform-state threshold (0.55). Both curves are essentially independent of system size (centre panel, shown for the neutral platform), as expected for a finite-horizon crossover rather than a phase transition. The similarity platform grows smoothly along λ toward a partial radicalization, with no crossover anywhere in the physical range—as the exact quadrature requires, since $t_{\mathrm{sim}} \simeq 507.7/\lambda$ exceeds the horizon $T = 80$ even at $\lambda = 1$; a weak, non-monotone trend with N (right panel), at the edge of statistical resolution, is consistent with a partial averaging-out of slot-level fluctuations—with only $k = 10$ slots, occasional high-influence cross-bloc sources produce bursts of repulsion that the population-wide weighting of Eq. (15) smooths away, which is also why the well-mixed curve (solid blue)

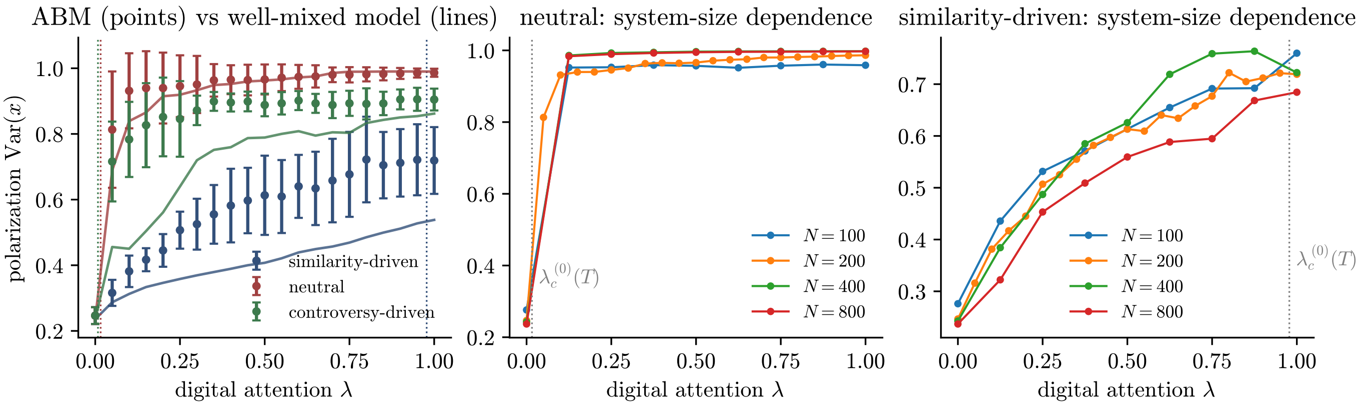

Figure 9: Radicalization onset in the digital attention share. Left: final polarization versus λ for the three platforms; points are the spatial ABM ($N = 200$, 12 seeds), solid lines the well-mixed particle model of Eq. (15) (12 realisations), dotted verticals the local-rate crossover estimates $\lambda_c^{(0)}(T)$ of Table 1, shown where they fall inside the sampled range (no parameters fitted to these data; the exact boundary-reaching crossover for the similarity kernel lies outside the physical interval, Sec. 3.2). Centre and right: system-size dependence of the $\mathrm{Var}(x)$–λ curves for the neutral and similarity platforms.

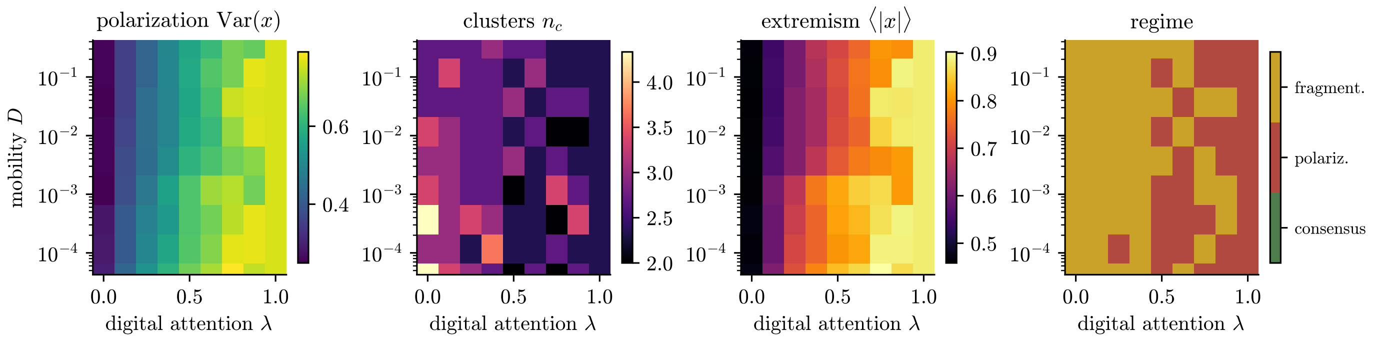

Figure 10: Phase diagram of the full Level-4 model (similarity-driven engagement, repulsion, heavy-tailed influence) in the mobility–digital-attention plane. From left to right: polarization $\mathrm{Var}(x)$, cluster count n_c, extremism $\langle|x|\rangle$, and the majority categorical state over 3 realisations per grid point. The panel is a coarse regime map intended to reveal qualitative structure; at 3 realisations per point the boundaries between regimes are not sharply resolved. At $\lambda = 1$ the physical layer is off and the dynamics is independent of D by construction.

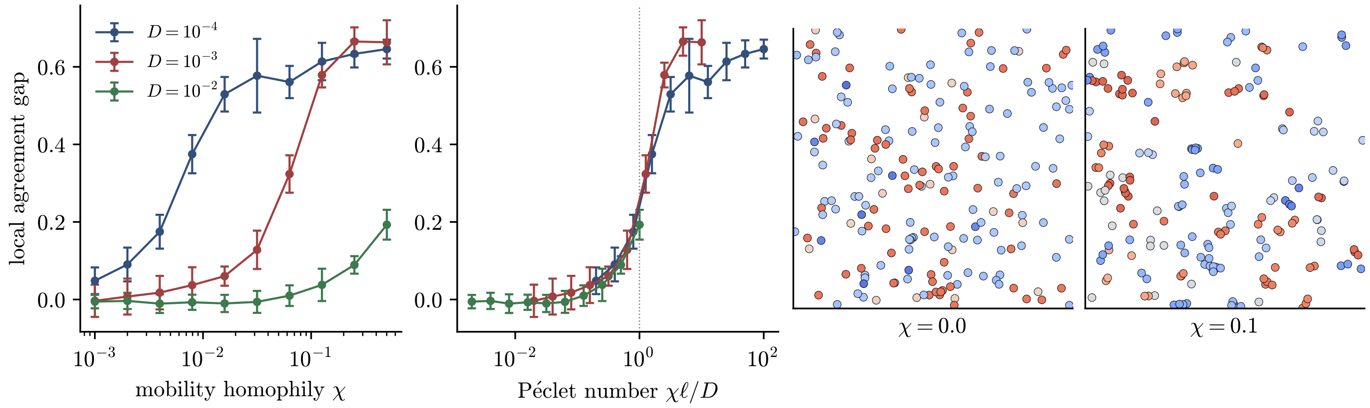

Figure 11: Level 5: homophilic mobility restores geographic opinion structure. Left: local agreement gap versus the homophilic drift strength χ for three mobilities. Centre-left: the same data against the Péclet number $\chi\ell/D$; the curves approximately collapse, with onset at $\mathrm{Pe} \sim 1$ (dotted). Right: final spatial configurations at $D = 10^{-3}$ without ($\chi = 0$) and with ($\chi = 0.1$) homophilic mobility.

sits below the ABM points while tracking their shape. The well-mixed particle model (12 realisations) reproduces the neutral curve nearly quantitatively for $\lambda \gtrsim 0.2$ (within ≈ 0.03, and within ≈ 0.01 for $\lambda \gtrsim 0.45$; at the steep onset $\lambda = 0.05$–0.15 it lags by up to ≈ 0.12) and sits below the controversy curve at all λ—by up to ≈ 0.33 near the onset, narrowing to ≈ 0.04 at $\lambda = 1$, for the same slot-fluctuation reason—with no parameters fitted to these data.

4.6. Phase diagram: mobility dependence vanishes at high digital attention

Figure 10 assembles the global picture in the (D, λ) plane, computed for the similarity-driven design—the slowest to radicalize, so the boundaries below are conservative; the other designs saturate at far lower λ. At $\lambda \approx 0$ the physical layer decides the outcome and the Level-1 crossover is recovered: frozen local fragmentation at low D, the mean-field cluster set at high D. As λ grows the platform captures the attention budget, the repulsive channel activates through the residual cross-cutting exposure, and both polarization and extremism rise across all mobilities: the high-λ region is a bimodal, radicalised state essentially independent of D. Once the platform curates most of an agent's exposure, it no longer matters how fast people move: geography sets the outcome only for societies whose attention it still owns.

4.7. Geography: a null result and its repair (Levels 1 and 5)

The bottom panel of Fig. 2 contains a null result that qualifies a common intuition. Moran's I—and the local agreement gap—never departs from noise level, even in the frozen regime where interactions are strictly local. The first-order part of this statement is a symmetry: on the torus, the initial law and the dynamics with $\chi = 0$ are translation invariant, so $\mathbb{E}[x_i \mid \mathbf{r}_i = \mathbf{r}]$ is independent of $\mathbf{r}$ at all times—no deterministic opinion map can exist. Symmetry does not forbid pair-level structure (nearby agents interacted more), but the measured pair correlations also stay at noise level: the ensemble mean of Moran's I is compatible with zero at every D (largest excursion 2.2 standard errors among eleven mobilities at 6 seeds each, so the null would not detect a weak true correlation), with an across-seed spread of ≈ 0.13, and the local agreement gap remains at 0.018 ± 0.027 even in the frozen regime—an order of magnitude below the 0.4–0.7 signal that develops once opinion–position coupling is switched on (Fig. 11). Disagreeing agents interleave at the same locations, interacting with disjoint compatible subsets, because nothing couples opinion to position. Locality controls how many opinions survive but not where they live. Within this model, low mobility alone is therefore insufficient to generate geographic echo chambers; persistent spatial domains require opinion–position coupling strong enough to compete with diffusion.

Level 5 quantifies how much coupling is needed. With the homophilic drift of Eq. (1) ($\chi > 0$: approach compatible neighbours, avoid incompatible ones [24, 25]), spatial opinion domains appear (Fig. 11). The control parameter is the Péclet number comparing drift and diffusion over the interaction range,

$$\mathrm{Pe} = \frac{\chi \ell}{D}, \tag{17}$$

and the local agreement gap for different mobilities approximately collapses onto a single curve in Pe, rising from near noise level for $\mathrm{Pe} \lesssim 0.3$ to strong spatial segregation for $\mathrm{Pe} \gtrsim 3$. Geographic opinion structure therefore exhibits a Péclet-controlled crossover in the strength of opinion-dependent mobility: it is weak below $\mathrm{Pe} \lesssim 0.3$ and pronounced above $\mathrm{Pe} \gtrsim 3$ in the explored range. The Brownian model of Levels 1–4 has $\chi = 0$ and therefore provides the natural null baseline by construction.

5. Discussion

5.1. Summary

We combined bounded-confidence opinion dynamics, Brownian and homophilic mobility, and an algorithmically curated digital layer under a finite attention budget, and found that the interaction between platform design and influence psychology—not either alone—decides the collective outcome. With assimilative influence, the neutral platform is a defragmenting force and homophilic curation builds echo chambers; with the repulsive channel active, all three tested designs exhibit outward radicalizing drift, but with strongly different finite-horizon magnitudes. At the reference Level-4 attention share the neutral platform reaches the maximal polarization permitted by the bounded opinion space, controversy-driven curation approaches a strong plateau, and homophilic curation delays the outward drift. The two-bloc reduction, the exact radicalization quadrature and its finite-horizon crossover, and the well-mixed comparison make the mechanism analytical rather than merely observational; the robustness and finite-size scans show that the inversion persists across the tested parameter and system-size ranges. A methodological implication follows: the naïve linear-stability analysis of the uniform state predicts thresholds an order of magnitude too large, because fragmentation preempts it—in coupled opinion–network systems, the state that matters for stability is the one the fast dynamics builds first.

5.2. Relation to experiments

The inversion offers a candidate population-scale mechanism consistent with the field experiment of Bail et al. [4], in which a month of bot-curated exposure to opposing views made Republican participants substantially more conservative (the corresponding shift for Democrats was not statistically significant—an asymmetry that our symmetric, one-dimensional model does not address): in our terms, an intervention that raises cross-cutting exposure moves a similarity-curated population toward the neutral level, and when the repulsive channel dominates, that movement increases polarization. We stress that this comparison is qualitative—no model parameter is calibrated to data—and that the repulsive channel itself is empirically contested (Sec. 2): the supporting longitudinal evidence [9] concerns a single non-political preference item (adolescents' taste in clothing style), finds little support for bounded confidence itself, and implies only low macro-level polarization in its own calibrated simulations, while controlled experiments find little evidence of negative shifts at all [32]. The field-experimental record on exposure interventions is equally mixed: counter-attitudinal exposure has also reduced affective polarization [5], and large platform experiments altering feed composition found little attitudinal effect [6]. Read this way, the model turns a psychological question into a measurable platform-policy question: it predicts intervention effects of either sign depending on the prevalence of negative influence in the target population—consistent with, though certainly not established by, the heterogeneous field record. When influence is assimilative, added exposure diversity heals fragmentation (Sec. 4.1); interventions on recommendation algorithms that ignore which influence regime the population is in can therefore backfire in either direction. At the collective level, our neutral-platform result reaches, by a different microscopic route—negative influence on a single opinion dimension rather than partisan sorting across many identity dimensions—a related qualitative conclusion to Törnberg [34]: digital interaction can increase polarization through exposure beyond local or like-minded social neighborhoods rather than through isolation alone. Recent voter-type models with involvement likewise find polarization rising with connectivity itself [13]; our mechanism differs by identifying repulsive cross-bloc exposure under a conserved attention budget as the kinetic control parameter.

5.3. The geography null result

Under opinion-independent mobility we detect no geographic opinion structure to destroy, within the resolution of our simulations: position and opinion stay uncorrelated at every mobility, by symmetry at first order and at pair level within the stated statistical power. Within this model, robust geographic opinion structure therefore requires an opinion–position coupling—homophilic migration, socially structured mobility, or spatially correlated exogenous influence—and Level 5 shows the coupling need only cross a Péclet-scale crossover centred around $\chi\ell/D \sim 1$. This sharpens, rather than contradicts, the mobile-agent literature [21, 24]: what matters is not whether agents move, but whether their movement knows about opinions.

5.4. Limitations and outlook

Opinions are one-dimensional and confidence bounds static; adaptive-confidence extensions are natural [31]. Opinions also live on a bounded interval, so "radicalization" here means pinning at the boundary of opinion space; in unbounded formulations [3] the analogue is unbounded divergence. The inversion mechanism—the cross-bloc exposure rate $p(y)$—does not depend on this choice, but the saturation values $\mathrm{Var}(x) \approx 1$ do. The two-bloc reduction is heuristic—it assumes symmetric blocs and fast rewiring—and its quantitative success invites a rigorous analysis of the well-mixed dynamics (15)—its McKean–Vlasov limit for $\kappa > 3$, the non-self-averaging regime at heavier tails, and the metastability structure behind the finite-horizon crossover [27, 28]. The attention budget is a fixed split rather than a dynamic allocation; letting λ itself be optimised by an engagement-maximising platform closes an obvious loop. Stubborn agents are implemented but unexplored; sweeping their number and placement against platform designs connects to opinion control [30]. The model also equates the engagement kernel with what agents attend to and are influenced by; real engagement optimisation responds to behaviour rather than opinion distance, and exposure need not translate into influence. The quantitative platform ranking is therefore a finite-time statement at the explored parameters and horizons, not a universal asymptotic order. Within the deterministic two-bloc reduction, any design with $p(y) > 0$ throughout the traversed range has the same boundary-pinned asymptotic destination, although the associated timescale can become arbitrarily long. Moreover, when two exposure profiles cross—as the controversy profile does with the neutral baseline—their relative ranking depends on the chosen target and observation horizon through Eq. (10). Finally, the model's sharpest empirical claim—that exposure diversity helps or harms depending on the prevalence of negative influence—is testable in principle by conditioning field-experiment outcomes on measured latitudes of acceptance.

Data and code availability

The full simulation code, the scripts that generate every figure and table, the raw sweep data, and animations of the coupled dynamics are openly available at https://github.com/REsteche/social-modeling.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation and revision of this work, the author used OpenAI ChatGPT and AI-assisted coding tools to support literature cross-checking, manuscript organization, language refinement, and consistency checking of analytical and numerical arguments. All model definitions, derivations, simulations, analyses, interpretations, references, and final manuscript content were independently reviewed and verified by the author, who takes full responsibility for the content of the article.

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